Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Introduction to Partial Differentiation

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Introduction to Partial Differentiation

1 Introduction to Partial Differentiation

Study focus: Introduction to Partial Differentiation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Anita from Department of Science and Humanities.

Key details

- As the name itself suggests partial differentiation is differentiating something partially.
- Maxima and minima for functions of two variables, Lagrange's method of undetermined multipliers, and as an application of all the Ebo concepts you will be learning errors and approximations.
- I know most of you might not like math that much, right?
- The only reason that you're here is maybe by force or you want to I hardly doubt that many of you want to learn things by choice, right?
- To make things a little more interesting, whenever we teach a topic, it's very essential that we ask ourselves questions.
- That is, why I need to study the topic, where exactly I'm going to apply the topic, and how can I apply the topic.
- How do we calculate the limit, continuity, and differentiability for a function of more than one variable?
- When we find the derivative, how can I apply that to find the relative extrema of functions?
- By the end of this unit, you will get the answers to all these questions.
- If we write like this, we know that y is a dependent variable which depends on x.
- But now we want to define a function which is a function of two independent variables.
- This is called a bivariate.
- Because both x and y are independent, we'll assign this function to another variable called z.
- whatever we do in this unit, we'll restrict ourselves to a bivariate function, but same thing can be extended to a multivariate function.

- The first case would be when you find the area of a rectangle.
- that area will be a function of just length and width.
- Electric potential if you want to find in within a cathode ratio, we know that it's a multivariable function with spatial coordinates as the variables as well as time t.
- The propagation of electromagnetic wave, which depends on the magnetic field, electric field, wavelength, and the direction of propagation.
- As you observe, this is a function of four independent variables.
- Not just that.
- you would need some parameters before you say what could be the weather in the next twenty four hours, right?
- But if there is a cyclone or any other major change in the weather, then you might have to include other parameters to decide the or to predict the weather conditions.

Example or application from the lecture

- On an larger scale, we can define a multivariable question as a function as quantities which depends on two or more variables are represented using functions of two or more variables, for example.
- Suppose you have a function of single variable.
- Let's say you have a function y is equal to f of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

the area, the population in that area, awareness among the people, the immunity level of the people who are living in that area, and the distribution of population based on the age group, gender, and other classifications.

Key details

- As you see, the Ebo examples make one thing clear that we are surrounded by function of more than one variable.
- A function of two variables denoted by so here we are denoting it as z is equal to f of x y.
- is a rule that assigns to each ordered pair of real numbers XY in a set D a unique real number denoted by f of x y.
- The set D is the domain of f and its range is the set of all the values that it takes.
- When do we say that this is a function you will have a domain which is a set of all real numbers.
- And then you have another set of real numbers called the range.
- And then we will map every element in the domain to a unique element in the range by this function f.
- But instead of having a real number here, I'll have an ordered pair here.
- When I put x is equal to 2 and y is equal to 1 in this expression, this will be mapped on to 8.
- The next question is, how does a function of more than one variable look like?

- It's just a parable which is symmetric about the y axis.
- But now if I ask you to plot a function z is equal to say sin x into cos y, it you can observe that it's not that easy for us to visualize how it looks.
- So we have luckily we have tools now which will help us to plot functions of n variables, at least two variables, because as I in increase the number of independent variables, the coordinate system has to increase.
- For a function with three independent variables, I have to add another dimension, which is practically not possible for us to demonstrate.
- Now this is the graph, the first graph that you see is for the function sin x into cos y.
- And this graph here is for a bivariate function again, which looks like e to the power of 1 minus x square minus y square.
- How does the graph of the multivariable function look like?
- The next thing would be we'll see how to find the limit of a function of two independent variables.
- Then we say that limit of f of x will go to a value L.

Example or application from the lecture

- But if you want to conclude or make a decision or draw a conclusion about any given situation, we have to consider always a function of more than one variable.
- What I mean by this in simple terms is suppose you have a function z is equal to f of x of single variable.
- For example, let's take suppose I have z is equal to x cube y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

As x approaches to A, if the only condition that I need to worry is as the distance between X and A reduces, the distance between f of x and l reduces.

Key details

- And then you calculate what happens to the value of the function.
- In both the cases, if it approaches to the value L, we say that the limit of the function exists and it is equal to L.
- We can do a similar thing for a function of two variables now.
- I say that the function f of x, comma y will tend to a limit L as.
- What do I mean by that?
- As I was mentioning, in case of single variable function, x will approach a either from the left or from the right.
- Let me choose a different color to show different path.
- It could be on a parabola.
- If we simplify this, we get 4 plus 4 minus 3 minus 8, this would be plus 3, and then minus 3 and then minus 6.
- I can cancel this, I can cancel this, and the final answer would be minus six.

- This is the graph of this function.
- How am I going to prove that it does not exist?

Example or application from the lecture

- In fact, what you did is suppose this is the line where I have my point A, then you can let X approach to A either from the left or from the right.
- X comma y approaches to a point, let's say x comma y approaches to the point, a comma b, if and only if this function f of x y takes the same value L.
- When xy approaches to a b along any given path.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

And that this line y is equal to 0 is nothing but x axis.

Key details

- In that case, what happens to the limit?
- I have to calculate limit as y is 0, I will get only x approaching to 0, 2x into 0 divided by 3x square plus 0.
- This will be anyway 0.
- What happens if it approaches along y is equal to x square?
- Wherever I have a y I need to replace it with x square, right?
- And if I simplify this, I'll get 2x cube divided by x square.
- If I take out x square common, I will get 3 plus x square.
- Of course I have limit.
- But what happens if I take the value that xy approaches to 00 along.
- The line y is equal to x.
- Wherever I have y, I'll replace it with x divided by 3x square plus x square.
- And if you observe carefully, this will be just 2x square divided by 4x square, and of course this is just half.
- That means the limit of this function does not exist.
- But it's fairly easier for us to check if the limit does not exist, because we can take two random paths and show that the value is not equal.
- We say that a function y is equal to f of x is continuous if it satisfies three properties.
- The first one is limit as extending to A of F of X must exist.
- The second thing would be f of a must exist.
- And the third one is limit as extending to A of F of X must be F of A.
- Similarly, in the next class we are going to define well how to define the continuity for a bivariate function or in general maybe for a multivariable function.
- So I'll see you all soon in the next class.

Example or application from the lecture

- Let's take along a different path.

- As you saw in the previous example, it won't be easy for us to check if the limit exists.
- Let's define now continuity for a function of a single variable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Introduction to Partial Differentiation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

C Revision and retrieval practice